

A Brief Primer on Transform Coding

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Abstract

We look at how transform coding works, its basic principles and a short look at the K-L transform: an "optimal" transform in terms of the least-square error sense.

1 Introduction

Lossy image compression techniques work in the spatial domain, since we are predicting the pixel values and the prediction errors (which are the values being encoded) also correspond to the pixel space. It is seen that the compression ratios of Differential Pulse Code Modulation (DPCM) encoded images are not always very high. This is primarily due to the fact that in presence of sharp changes in intensity values, which are always expected in any natural image, prediction suffers and encoding large prediction errors in those regions lead to high consumption of bits.

In transform-domain approaches, the pixel intensities are first mapped into a set of **linear, reversible** transform coefficients, which are subsequently quantized and encoded.

2 Transform Coding

The basic principle of transform coding is to map the pixel values into a set of linear transform coefficients, which are subsequently quantized and encoded. By applying an inverse transformation on the decoded transform coefficients, it is possible to reconstruct the image with some loss. It must be noted that *the loss is **not** due to the process of transformation and inverse transformation, but due to quantization alone.*

Since the details of an image and hence its spatial frequency content vary from one local region to the other, it is better to apply the transformation on local areas of the image, rather than applying a global transformation on the entire image.

For transform coding, the first and foremost step is to subdivide the image into non-overlapping blocks of fixed size. Without loss of generality, we can consider a square image of size $N \times N$ pixels and divide it into n^2 blocks, each of size $\frac{N}{n} \times \frac{N}{n}$, where $n \ll N$ and is a factor of N .

2.1 What kinds of transforms are allowable?

A transformation must necessarily fulfill the following properties –

1. The coefficients in the transformed space should be de-correlated.

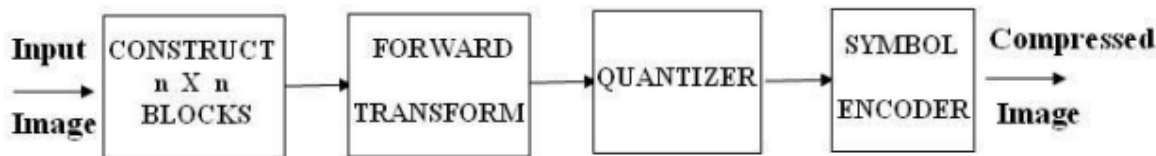


Figure 1: Block Diagram of Transform Coding System Encoder



Figure 2: Block Diagram of Transform Coding System Decoder

2. Only a limited number of transform coefficients should carry most of the signal energy (in other words, the transformation should possess energy compaction capabilities) and most of the coefficients should carry insignificant energy. Then the quantization process can coarsely quantize those "extra" coefficients to achieve compression, without much perceptible degradation.

2.2 Generalized forward and inverse transforms

Consider an image block of size $n \times n$ whose pixel intensities are represented as $s[n_1, n_2]$, with $n_1, n_2 = 0, 1, \dots, n-1$. Then the general expression for the transform coefficients is given as

$$S[k_1, k_2] = \sum_{n_1=0}^{n-1} \sum_{n_2=0}^{n-1} s[n_1, n_2] g(n_1, n_2, k_1, k_2) \quad (1)$$

for $k_1, k_2 = 0, 1, \dots, n-1$, where $g(n_1, n_2, k_1, k_2)$ is the transformation kernel.

Given the transformation coefficients, the original image can be regained as

$$s[n_1, n_2] = \sum_{k_1=0}^{n-1} \sum_{k_2=0}^{n-1} S[k_1, k_2] h(n_1, n_2, k_1, k_2) \quad (2)$$

for $n_1, n_2 = 0, 1, \dots, n-1$, where $h(n_1, n_2, k_1, k_2)$ is the inverse transform kernel.

2.3 Separable and Symmetric Transforms

A transformation kernel is said to be separable if it can be expressed as a product of two kernels along the row and the column:

$$g(n_1, n_2, k_1, k_2) = g_1(n_1, k_1) g_2(n_2, k_2)$$

Separable transforms are easier to implement in hardware, since the transformation can first be applied along the rows and then along the columns.

A separable transform is symmetric, if the kernels along the row and the column have the identical function, i.e. $g_1 = g_2$.

2.4 Basis Images

From 2 we have n^2 such equations, one for each pair (n_1, n_2) . These equations can be combined in matrix form to read as:

$$\mathbf{s} = \sum_{k_1=0}^{n-1} \sum_{k_2=0}^{n-1} S[k_1, k_2] \mathbf{H}_{k_1, k_2}$$

where

$$\mathbf{H}_{k_1, k_2} = \begin{bmatrix} h(0, 0, k_1, k_2) & h(0, 1, k_1, k_2) & \cdots & h(0, n-1, k_1, k_2) \\ h(1, 0, k_1, k_2) & \cdots & & \\ \vdots & \ddots & \cdots & \vdots \\ h(n-1, 0, k_1, k_2) & h(n-1, 1, k_1, k_2) & \cdots & h(n-1, n-1, k_1, k_2) \end{bmatrix}$$

The image block $\mathbf{s}$ can therefore be realized by a weighted summation of n^2 images, each of size $n \times n$, where the weights are provided by the transform coefficients $S(k_1, k_2)$. The matrix $\mathbf{H}_{k_1, k_2}$ is known as a basis image corresponding to (k_1, k_2) . There are n^2 such basis images.

3 The Covariance Matrix

A block may be represented by a n^2 - dimensional random variable vector $\mathbf{x}$, whose elements are composed by the lexicographic ordering of pixel intensity values. We define a vector $\mathbf{b}$, such that $\mathbf{b} = \mathbf{x} - \mathbb{E}[\mathbf{x}]$. We approximate $\mathbb{E}[\mathbf{x}] = \mu$, where $\mu = \frac{1}{N_B} \sum_i \mathbf{x}_i$ where N_B is the number of $n \times n$ blocks present.

The covariance matrix computed over blocks of size $n \times n$ is defined by

$$\mathbf{R}_b = \mathbb{E}[\mathbf{b}\mathbf{b}^T] = \mathbb{E}[(\mathbf{x} - \mu)(\mathbf{x} - \mu)^T]$$

where, as before, the expectation is approximated by averaging over all the blocks. Since $\mathbf{b}$ is an n^2 dimensional vector, its outer product realizes an $n^2 \times n^2$ - dimensional matrix, which is the size of $\mathbf{R}_b$. The matrix is real and symmetric and it is possible to find a set of n^2 orthonormal eigenvectors. (Why?)

Let $\mathbf{e}_i$ and λ_i , $i = 1, 2, \dots, n^2$ be the eigenvectors and corresponding eigenvalues such that the eigenvalues are arranged in non-increasing order such that $\lambda_j \geq \lambda_{j+1}$. Then, by the definition of eigenvectors and eigenvalues, we can write

$$\mathbf{R}_b \mathbf{e}_j = \lambda_j \mathbf{e}_j, \quad j = 1, 2, \dots, n^2$$

Since $\mathbf{e}_j^T \mathbf{e}_j = 1$ we can pre-multiply $\mathbf{e}_j^T$ to get

$$\mathbf{e}_j^T \mathbf{R}_b \mathbf{e}_j = \lambda_j \quad (3)$$

Now compose a matrix of Γ dimension $n^2 \times n^2$ such that its rows are formed from the eigenvectors of $\mathbf{R}_b$, ordered such that the first row corresponds to the eigenvector corresponding to the largest eigenvalue and the corresponding values of the eigenvalues decreases thereafter. Then, we can write 3 in matrix form as:

$$\Gamma \mathbf{R}_b \Gamma^T = \Lambda \quad (4)$$

, where Λ is the diagonal matrix of ordered eigenvalues (why this "ordering" is necessary will be revealed soon).

Noting that by the orthonormal properties of Γ , we get $\Gamma^T = \Gamma^{-1}$ and so we obtain

$$\mathbf{R}_b = \Gamma^T \Lambda \Gamma$$

4 Using the covariance matrix: The Karhunen-Loeve (K-L) Transform

We intend to use the matrix Γ to map the n^2 -dimensional vector $\mathbf{b}$ to an n^2 -dimensional vector $\mathbf{y}$, defined by $\mathbf{y} = \Gamma \mathbf{b}$. This is the **K-L Transform**.

The covariance matrix $\mathbf{R}_y$ of the $\mathbf{y}$ vectors is given as

$$\mathbf{R}_y = \mathbb{E}[\mathbf{y}\mathbf{y}^T] = \mathbb{E}[(\Gamma \mathbf{b})(\Gamma \mathbf{b})^T] = \Gamma \mathbb{E}[\mathbf{b}\mathbf{b}^T] \Gamma^T = \Gamma \mathbf{R}_b \Gamma^T = \Lambda \quad (\text{by (4)})$$

The covariance matrix has the same eigenvectors and eigenvalues as that of $\mathbf{R}_b$, but its off-diagonal elements are zero, which signifies that **the elements of the transform-domain vectors $\mathbf{y}$ are uncorrelated**, which is what we want from the transform-domain coefficients.

The vector $\mathbf{b}$ can be recovered as

$$\mathbf{b} = \Gamma^{-1} \mathbf{y}$$

and hence the original block is reconstructed using $\mathbf{x} = \Gamma^{-1} \mathbf{y} + \mu$.

Suppose that instead of using all the n^2 eigenvectors of $\mathbf{R}_b$, we use only k eigenvectors corresponding to the k largest eigenvalues and form a transformation matrix Γ_k of order $k \times n^2$. The resulting transformed vector $\hat{\mathbf{y}}$ therefore becomes k -dimensional and the reconstruction $\hat{\mathbf{x}} = \Gamma_k^T \hat{\mathbf{y}} + \mu$ will not be exact.

4.1 Optimality of the K-L transform

If we project the mean-removed vector $\mathbf{b}$ into any of the eigenvectors $\mathbf{e}_j$, the projection is defined by the inner product of the vectors $\mathbf{b}$ and $\mathbf{e}_j$, and is given by

$$A = \mathbf{b}^T \mathbf{e}_j = \mathbf{e}_j^T \mathbf{b}$$

The variance σ^2 of the projection is therefore given by

$$\sigma^2 = \mathbb{E}[A^2] = \mathbb{E}[\mathbf{e}_j^T \mathbf{b} \mathbf{b}^T \mathbf{e}_j] = \mathbf{e}_j^T \mathbb{E}[\mathbf{b} \mathbf{b}^T] \mathbf{e}_j = \mathbf{e}_j^T \mathbf{R}_b \mathbf{e}_j = \lambda_j$$

By projecting the vector $\mathbf{b}$ into all the n^2 eigenvectors, we obtain the total variance as $\sigma_{n^2}^2 = \sum_{j=1}^{n^2} \lambda_j$. By considering only the first k eigenvectors, the variance of the approximating signal in the projected space is given by $\sigma_k^2 = \sum_{j=1}^k \lambda_j$ (which is where the ordering comes into play, as this is the choice of the k largest eigenvalues).

Thus, the mean-square error in the projected space by considering only the first k components can be obtained as

$$e_{ms} = \sigma_{n^2}^2 - \sigma_k^2 = \sum_{j=k+1}^{n^2} \lambda_j$$

Since, the transformation is energy-preserving, the same mean-square error exists between the original vector $\mathbf{x}$ and its approximation $\hat{\mathbf{x}}$. Since the λ_j s decrease monotonically, we can minimize this error by choosing the largest k eigenvalues.

Thus, K-L transform is optimal in the sense that it minimizes the mean-square error between the original input vectors $\mathbf{x}$ and their approximations $\hat{\mathbf{x}}$.